

Supplementary Material: AGILE: Hand-Object Interaction Reconstruction from Video via Agentic Generation

JIN-CHUAN SHI*, State Key Lab of CAD & CG, Zhejiang University, China

BINHONG YE*, State Key Lab of CAD & CG, Zhejiang University, China

TAO LIU, State Key Lab of CAD & CG, Zhejiang University, China

XIAOYANG LIU, State Key Lab of CAD & CG, Zhejiang University, China

YANGJINHUI XU, State Key Lab of CAD & CG, Zhejiang University, China

JUNZHE HE, State Key Lab of CAD & CG, Zhejiang University, China

ZEJU LI, State Key Lab of CAD & CG, Zhejiang University, China

HAO CHEN[†], State Key Lab of CAD & CG, Zhejiang University, China

CHUNHUA SHEN, State Key Lab of CAD & CG, Zhejiang University, China and Zhejiang University of Technology, China

CCS Concepts: • **Computing methodologies** → **Reconstruction**; *Mesh geometry models*.

ACM Reference Format:

Jin-Chuan Shi, Binhong Ye, Tao Liu, Xiaoyang Liu, Yangjinhui Xu, Junzhe He, Zeju Li, Hao Chen, and Chunhua Shen. 2026. Supplementary Material: AGILE: Hand-Object Interaction Reconstruction from Video via Agentic Generation. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers (SIGGRAPH Conference Papers '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3799902.3811134>

1 Implementation Details

Pipeline and dependencies. Our framework integrates several state-of-the-art foundation models. Crucially, we utilize **Gemini 3 Pro** [Comanici et al. 2025] as the VLM agent responsible for keyframe selection and rigorous quality assessment. Guided by this agent, multi-view image synthesis is performed using the Gemini 2.5 Flash image generation model [Comanici et al. 2025], while subsequent 3D mesh and texture generation are handled by Hunyuan 3D [Team 2025a,b]. For preprocessing, geometric priors including metric depth and segmentation masks are extracted via MoGe-2 [Wang et al. 2025] and SAM2 [Ravi et al. 2024], respectively, with the initial object pose estimated by FoundationPose [Wen et al. 2024]. All differentiable

*Equal contribution.

[†]Corresponding author.

Authors' Contact Information: Jin-Chuan Shi, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Binhong Ye, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Tao Liu, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Xiaoyang Liu, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Yangjinhui Xu, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Junzhe He, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Zeju Li, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Hao Chen, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China; Chunhua Shen, State Key Lab of CAD & CG, Zhejiang University, Hangzhou, China and Zhejiang University of Technology, Hangzhou, China.



This work is licensed under a Creative Commons Attribution 4.0 International License.

SIGGRAPH Conference Papers '26, Los Angeles, CA, USA

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2554-8/2026/07

<https://doi.org/10.1145/3799902.3811134>

rendering is implemented using PyTorch3D [Ravi et al. 2020]. The entire pipeline, from preprocessing to optimization, is designed to run efficiently on a single NVIDIA 4090 GPU.

Automated retopology and UV unwrapping. After the initial 3D mesh is generated by Hunyuan 3D, we apply automated retopology and UV unwrapping to produce a clean, simulation-ready asset. Specifically, we leverage Blender's built-in quad-remeshing algorithm to convert the raw triangular mesh into a uniform quad-dominant topology with a target face count of approximately 10,000 faces. This step ensures the mesh has a regular, artifact-free topology suitable for physics simulation and collision detection. Subsequently, we perform automatic UV projection using Blender's Smart UV Project, which segments the surface into non-overlapping UV islands and minimizes distortion. The original texture is then baked onto the new UV layout via ray-cast projection, preserving the high-frequency surface details from the generative model. This fully automated process eliminates the need for any manual post-processing.

Online optimization strategy. Our method optimizes each frame sequentially via a two-stage alternating approach to ensure stability. Stage one optimizes hand translation (up to 200 iterations), followed by object 6D pose refinement (up to 400 iterations) with fixed hand parameters. exclusively for the onset frame, we optimize object scale by minimizing $\mathcal{L}_{\text{mask}}$ and the ray-traced contact loss $\mathcal{L}_{\text{contact}}$. During scale optimization, object rotation is fixed, and translation is constrained to the visual ray passing through the object center, thereby distinctively regularizing the solution space. Convergence is reached when the moving average of parameter variations, including scale when active, is less than 10^{-4} .

Optimizers and learning-rate schedule. For optimization, we employ separate Adam optimizers [Kingma 2014] for rotation and translation parameters. The object optimization uses fixed learning rates of 2×10^{-3} for rotation and 1×10^{-3} for translation. Similarly, hand optimization uses fixed learning rates of 1×10^{-3} for translation. We adopt constant learning rates throughout optimization, as we empirically observe that learning-rate decay does not lead to

measurable accuracy improvements in our setting, while increasing the number of iterations required for convergence.

Loss terms and weights. The loss function employs the following weights: $\mathcal{L}_{\text{mask}}$ (L2) is weighted at 5.0, $\mathcal{L}_{\text{dino}}$ (L1) at 10.0. For physical interaction modeling, the ray-traced contact loss $\mathcal{L}_{\text{contact}}$ receives a weight of 5.0. Additionally, the relative position interaction loss $\mathcal{L}_{\text{interact}}$ is weighted at 400 for HO3D dataset and 200 for DexYCB dataset, with a maximum distance threshold of 0.05m to prevent spurious long-range attractions between hand and object. For the SDF-based distance-weight gating function $w_i = 1 - \tanh(\sigma \cdot d_i)$ defined in the main paper, we set $\sigma = 40$.

Interaction Onset Frame selection. For HO3D, we identify the interaction onset as the earliest frame where the object exhibits significant motion while remaining fully visible. We quantify the motion by measuring the variation of the non-occluded object mask between consecutive frames:

$$r_i = \frac{\|(\mathcal{M}_o^{i+1} - \mathcal{M}_o^i) \odot (1 - \mathcal{M}_h^{i+1}) \odot (1 - \mathcal{M}_h^i)\|_1}{\|\mathcal{M}_o^i\|_1 + \varepsilon}, \quad (1)$$

where $\mathcal{M}_o$ and $\mathcal{M}_h$ denote the object and hand masks, respectively. The numerator captures the change in visible object pixels, while the denominator normalizes by the object area ($\varepsilon = 10^{-8}$ ensures stability). We select the first frame i that satisfies two criteria: (1) the motion ratio r_i exceeds a threshold $\tau = 0.025$, and (2) the object mask does not touch the image boundaries. The latter ensures the object is not truncated, thereby guaranteeing reliable geometric supervision.

2 Why Start at Interaction Onset?

We choose the interaction onset frame as the optimization anchor for two key reasons. First, metric scale alignment: since monocular object reconstruction suffers from scale ambiguity, the physical contact allows us to leverage the hand’s reliable metric scale to constrain and propagate the correct object size. Second, loss efficacy: the contact-based losses ($\mathcal{L}_{\text{contact}}$) are physically meaningful only when interaction occurs. Initiating optimization at this frame ensures these geometric constraints provide valid gradients for precise hand-object alignment.

3 Clarification on the Static Camera Assumption

We clarify that the static camera assumption in our framework is strictly limited to the automatic selection of the Interaction Onset Frame (IOF), rather than the core reconstruction pipeline itself.

Why a static camera for IOF detection? As described in Sec. 1, we identify the IOF by measuring the 2D pixel displacement of the unoccluded object mask between consecutive frames. This specific heuristic assumes a fixed camera to reliably distinguish genuine object movement from camera ego-motion. Under a moving camera, apparent mask displacement could arise from viewpoint change rather than object manipulation, leading to incorrect IOF identification.

Core pipeline independence. Crucially, our core pipeline—Agentic Generation (Sec. 3.1), Pose Initialization (Sec. 3.2), and Contact-Aware Tracking (Sec. 3.3)—operates entirely in the camera coordinate system. Since all optimization variables (hand pose, object pose, and interaction constraints) are defined relative to the camera frame, camera ego-motion is mathematically irrelevant to our optimization. This means the reconstruction quality is not affected by whether the camera is static or dynamic, as long as a valid IOF is identified.

Extension to dynamic cameras. To handle dynamic cameras (e.g., first-person or egocentric videos), one simply needs to substitute the IOF selection step with alternative methods—such as VLM-based action recognition (e.g., detecting the moment of grasping), off-the-shelf contact predictors, or manual specification. Camera intrinsics and extrinsics for each frame can be recovered using existing structure-from-motion or SLAM pipelines (e.g., MegaSaM [Li et al. 2025]). The rest of our framework requires absolutely zero modification. We view this modularity as a strength: the IOF selection is a replaceable preprocessing component, while the core method is camera-motion agnostic.

4 Details of Fine-Grained Ablation Variants

The fine-grained ablation results on Agentic Multi-view Generation are reported in Table 3 of the main text (variants b_1 , b_2). Here we describe the ablation configurations in detail:

(1) *Heuristic Selection* (b_1): We replace the VLM-based keyframe selection with a heuristic that simply chooses the frames with the largest object mask areas. Multi-view synthesis and all subsequent steps remain unchanged. This isolates the contribution of the VLM’s spatial reasoning for selecting geometrically informative views.

(2) *w/o Multi-View Synthesis* (b_2): We retain VLM keyframe selection but skip the multi-view synthesis stage, directly feeding the single selected keyframe to the 3D generation model. This isolates the contribution of multi-view consistency for faithful 3D reconstruction.

Both variants use the *exact same* foundation models (Gemini, Hunyuan3D) as the full pipeline to ensure a fair comparison.

5 Sequence Used

We evaluate our method on sequences from the DexYCB [Chao et al. 2021] and HO3D [Hampali et al. 2020] datasets. Specifically, as shown in Table 1 and Table 2, we randomly select 18 sequences from HO3D and 20 sequences from DexYCB, covering a diverse range of object types and hand-object interaction patterns. For the HO3D dataset, we begin processing from the identified IOF to ensure meaningful interaction modeling. HO3D sequences vary in length from 800 to 1600 frames; for computational efficiency, we cap the processing at 1024 frames per sequence. For the DexYCB dataset, each selected sequence spans approximately 80 frames. Unless otherwise specified, we uniformly subsample all sequences with a stride of 5 (i.e., processing every fifth frame), which balances temporal coverage with computational cost. These sequences are chosen to provide comprehensive coverage of common manipulation scenarios and object categories, enabling robust evaluation of our method’s generalization capabilities across different interaction types and geometric configurations.

Table 1. Sequences selected from the HO3D datasets.

Sequence ID	Object	IOF Index
ABF12	bleach cleanser	100
ABF14	bleach cleanser	100
BB12	banana	200
BB13	banana	130
GPMF12	potted meat	105
GPMF14	potted meat	100
GSF13	scissors	100
GSF14	scissors	80
MC1	cracker box	0
MC4	cracker box	0
MDF12	power drill	150
MDF14	power drill	250
ShSu10	sugar box	50
ShSu12	sugar box	110
SM2	mustard	0
SM4	mustard	0
SMu1	mug	20
SMu40	mug	0

Table 2. Sequences selected from the DexYCB datasets.

Sequence ID	Object
20200709-subject-01/20200709_141754/836212060125	master chef can
20200709-subject-01/20200709_142853/836212060125	master chef can
20200813-subject-02/20200813_154204/836212060125	bleach cleanser
20200813-subject-02/20200813_153453/836212060125	power drill
20200820-subject-03/20200820_141856/836212060125	potted meat
20200820-subject-03/20200820_143330/836212060125	mug
20200903-subject-04/20200903_103554/836212060125	master chef can
20200903-subject-04/20200903_113012/836212060125	wood block
20200908-subject-05/20200908_144138/836212060125	tomato soup can
20200908-subject-05/20200908_152416/836212060125	large clamp
20200918-subject-06/20200918_113441/836212060125	sugar box
20200918-subject-06/20200918_114747/836212060125	potted meat
20200928-subject-07/20200928_153800/836212060125	pitcher base
20200928-subject-07/20200928_144300/836212060125	mustard bottle
20201002-subject-08/20201002_110940/836212060125	pitcher base
20201002-subject-08/20201002_105558/836212060125	mustard bottle
20201015-subject-09/20201015_143700/836212060125	mustard bottle
20201015-subject-09/20201015_142940/836212060125	cracker box
20201022-subject-10/20201022_111745/836212060125	mustard bottle
20201022-subject-10/20201022_111209/836212060125	sugar box

6 Analysis on Results of Baselines

On the DexYCB dataset, both baselines exhibited varying degrees of failure. Specifically, HOLD encountered issues in some sequences where it failed to obtain the hand/object mesh due to inaccurate poses, which prevented the geometric structure from being effectively learned. Meanwhile, MagicHOI was unable to complete colmap reconstruction in most sequences and thus could not obtain stable poses for subsequent processing. On the other hand, MagicHOI’s hand-object alignment relies on the geometry generated by its generative model. Due to the limitations of its capabilities, it fails

Table 3. Comparison of computational cost for a 200-frame sequence on a single RTX 4090 GPU. Our method requires significantly less preprocessing and total runtime than existing methods.

Method	Preprocessing	Optimization	Total
HOLD	~65 min	~24 h	~25.1 h
MagicHOI	~25 min	~2 h	~2.4 h
Ours	~21–29 min	~1.75 h	~2.1–2.2 h

to produce reasonable geometric structures in some challenging cases, leading to the failure of the final results.

7 Computation Cost

Our optimization process is computationally efficient. On a single NVIDIA RTX 4090 GPU, each frame requires approximately 30–50 seconds for optimization. Unless otherwise specified, we process every fifth frame of each sequence. As a result, the total computation time scales linearly with the sequence length. On average, a sequence of 1000 frames (200 frames after downsampling) can be processed in approximately 1.5 hours, which is consistent with the overall runtime reported in Table 3.

Detailed runtime breakdown. We report the end-to-end runtime for a representative 200-frame sequence (1000 raw frames at stride 5) on a single NVIDIA RTX 4090 GPU. Despite introducing a VLM critic in the generation loop, our total pipeline time remains competitive with or faster than existing baselines:

- **HOLD:** ~25.1 hours (Preprocessing: ~65 min; Optimization: ~24 h)
- **MagicHOI:** ~2.4 hours (Preprocessing: ~25 min; Optimization: ~2 h)
- **Ours:** ~2.1–2.2 hours (Preprocessing: ~21–29 min; Optimization: ~1.75 h)

Breakdown of our preprocessing (~21–29 min): The Agentic Generation stage accounts for ~16–24 minutes, depending on the number of rejection sampling iterations ($n \in [1, 3]$). This includes keyframe selection (~2 min), 3D generation (~10 min), multi-view synthesis (~2–6 min), and texture refinement (~2–6 min). The remaining preprocessing takes ~5 minutes: mask extraction via SAM2 (~20 s), metric depth via MoGe-2 (~88 s), hand pose via WiLoR (~24 s), and FoundationPose initialization (~160 s).

8 Details of VLM-Guided 3D Generation

In this section, we provide the detailed prompt specifications and evaluation protocols used in our agentic generation pipeline. Our framework leverages a Vision-Language Model (VLM) as an intelligent supervisor to guide three critical stages: (1) informative keyframe selection, (2) consistent multi-view synthesis, and (3) high-fidelity texture refinement. For each stage, we present the system instructions, task definitions, and representative VLM responses. The prompts presented below are the exact instructions used in our pipeline to ensure reproducibility.

8.1 Stage 1: Informative Keyframe Selection

The objective of this stage is to filter the input video stream and identify a minimal set of frames that maximize geometric coverage while minimizing occlusion.

8.1.1 Prompt Specification.

```
{
  "system_role": "You are an expert in computer
    vision, specializing in selecting frames that
    best showcase the complete appearance of an
    object from multiple images.",
  "task": "Select {num_frames} images from the
    following candidate frames. The selected images
    must:
    1. Clearly display different angles of the object.
    2. Maximize coverage of the object's complete
    appearance (front, back, left, right, top,
    bottom).
    3. Be sharp, with the object fully visible and
    minimal occlusion.
    4. Have the maximum possible viewpoint difference
    between selected frames.
    5. Feature the object occupying a relatively large
    portion of the frame.",
  "response_format": "
  {
    "selected_frames": [1, 5, 10, 15], // Indices of
    selected frames
    "reasoning": "Reason for selection",
    "coverage": {
      "front": true,
      "back": true,
      "left": true,
      "right": false
    }
  },
  "instruction": "Please reply strictly in JSON
    format without adding other explanatory text."
}
```

Listing 1. Prompt for Keyframe Selection

8.1.2 Example Response.

```
{
  "selected_frames": [1, 5, 13, 21],
```

```
  "reasoning": "Frame 1 is selected as the standard
    front view, clearly displaying the main brand
    label. Frame 5 shows the back of the object,
    clearly presenting the ingredients list and
    nutritional information. Frame 13 provides the
    side profile, effectively showcasing the object
    's depth, thickness, and side curves. Frame 21
    captures the bottom perspective, supplementing
    this often-overlooked angle. All four images
    are high-definition, have a large object-to-
    frame ratio, and exhibit significant viewpoint
    differences, achieving maximum multi-angle
    coverage.",
  "coverage": {
    "front": true,
    "back": true,
    "left": true,
    "right": false,
    "top": false,
    "bottom": true
  }
}
```

Listing 2. VLM Response for Keyframe Selection

8.2 Stage 2: Multi-View Synthesis and Validation

Based on the selected keyframes, we generate four orthogonal views. To ensure consistency, a VLM critic evaluates the generated views against strict geometric and texture criteria.

8.2.1 Generation Prompts.

```
{
  "multiview_generation": {
    "static_object": "Generate a complete image of the
      water bottle in the picture from four
      perspectives: front, back, left, and right.
      Require photorealism. Present as a four-view
      grid. Remove the plate and extraneous objects.
      Pure white background. The image must not
      contain text indicating viewpoint direction (e.
      g., front, back, left, right).",
    "hand_held_object": "Generate a complete image of
      the object held in the hand from four
      perspectives: front, back, left, and right.
      Require photorealism. Present as a four-view
      grid. Remove the hand and hallucinate/complete
      the regions occluded by the hand. Pure white
      background. The image must not contain text
      indicating viewpoint direction. The object size
      should be consistent across all four views."
  }
}
```

Listing 3. Prompts for Multi-View Generation

8.2.2 Validation Protocol (The "Critic").

```
{
  "validation": {
    "system_role": "You are an expert in 3D modeling,
      material analysis, and multi-view image quality
      assessment. Please carefully analyze the
      following images:",
    "task_description": "Task:
      1. The first image(s) are the original input,
        showing the appearance, texture, and material
        of an object.
      2. The last image is the generated 'four-view'
        image, which should display the complete image
        of the object from four different perspectives
        (front, back, left, right) while preserving the
        original visual attributes.",
    "evaluation_criteria": "Criteria:

    Level 1: Veto Items
    1. Text Check: Does the generated image contain
      any text, labels, or viewpoint descriptions (e.
      g., front, back)? If yes, terminate evaluation;
      result is invalid.

    Level 2: Core Dimension Scoring (0-10)
    2. Geometry & View Correctness (Weight: 30%): Are
      viewpoints correct? Is orientation consistent (
      no rotation)? Any rotation results in large
      deductions.
    3. Texture & Material Fidelity (Weight: 20%): Are
      surface textures (e.g., patterns) and material
      properties (e.g., reflection) consistent with
      the original?
    4. Geometric Detail Integrity (Weight: 20%): Are
      key geometric details (chamfers, holes,
      embossing) preserved?
    5. Feature Consistency (Weight: 15%): Is it the
      same object in terms of shape, style, and color
      ?
    6. Image Quality (Weight: 15%): Is the image clear
      , noise-free, and on a pure white background?

    Level 3: Deductions
    - Rotated views: -3 points each.
    - Poor layout: -1 to -2 points.",
    "response_format": "JSON format containing:
      is_valid, score_overall, score_breakdown,
      has_text, rotated_views,
      improvement_suggestions, summary_feedback, etc.
      ",
    "instruction": "Reply strictly in JSON. First
      check veto items."
  }
}
```

Listing 4. Prompt for Multi-View Validation

8.3 Stage 3: Texture Refinement and Agentic Editing

After lifting the multi-view images to 3D and unwrapping UVs, we perform a texture refinement step. A VLM critic detects artifacts (e.g., hallucinated content, missing details) and guides a subsequent editing pass.

8.3.1 Texture Validation Protocol.

```
{
  "texture_validate": {
    "system_role": "You are an expert in 3D texture
      mapping analysis, material comparison, and
      image consistency assessment. Carefully analyze
      the images.",
    "task_description": "Task:
      1. Image 1: The four-view image.
      2. Image 2: The generated Texture Map.
      3. Evaluate if the texture map accurately
        reproduces all texture information from the
        four-view image, focusing on completeness,
        fictional content, and correspondence.",
    "evaluation_criteria": "Criteria:

    Level 1: Veto Items
    - Invalid texture map (blank, pure color, severe
      distortion).
    - Key features completely missing.

    Level 2: Core Dimensions
    1. Completeness (Weight: 30%): Are all visible
      textures present? (Deduction: -2 per major
      missing item).
    2. Accuracy/No Fiction (Weight: 25%): Does it
      contain hallucinated content not present in the
      source? (Severe penalty: -3 to -5 points).
    3. Correspondence (Weight: 20%): Are textures
      mapped to correct UV islands?
    4. Color/Material (Weight: 15%): Consistency in
      color, shading, and saturation.
    5. Fidelity (Weight: 10%): Resolution and detail
      preservation.

    Level 3: Extra Deductions
    - Seams, repetitions, stretching.",
    "response_format": "JSON format containing:
      is_valid, score_overall,
      fictional_content_analysis,
      missing_content_analysis,
      texture_coverage_analysis, etc.",
  }
}
```

```

"instruction": "Strictly follow JSON format. Pay
special attention to 'fictional content', any
texture appearing in the map but not in the
source views is a severe error!"
}
}

```

Listing 5. Prompt for Texture Map Validation

8.3.2 Example Validation Result (Invalid Case Detected).

```

{
  "is_valid": false,
  "score_overall": 0.0,
  "fictional_content_analysis": {
    "has_fictional_content": true,
    "fictional_items": [
      {
        "type": "severe",
        "description": "The entire back texture is a
fabricated copy of the front label instead of
the nutrition facts panel shown in the source.",
        "location_in_texture": "The second large body
island in the texture map.",
        "deduction": 5
      }
    ],
    "total_fictional_deduction": 5
  },
  "summary_feedback": "Invalid. Critical error: the
back side texture is a fabrication (copy of the
front). This makes the model unusable for pose
estimation from rear viewpoints."
}

```

Listing 6. Example Validation Result detecting hallucinations

8.3.3 Refinement Prompt Construction.

Based on the negative feedback from the validation step, we programmatically construct a structured refinement command to guide the image-to-image editing model.

```

{
  "texture_refinement_task": {
    "base_instruction": "Below are object images from
multiple views and their texture map. Please
edit the texture map based on the following
critique:",
    "injected_feedback": "Invalid. Critical error: the
back side texture is a fabrication (copy of
the front). This makes the model unusable for
pose estimation from rear viewpoints.",

```

```

"geometric_constraints": "Do not change the layout
or shape of UV islands. Only align the content
with reference images. The bottom-left area
corresponds to the bottom view.",
"specific_corrections": "Change the purple/black
artifact regions to match the red color of
other parts. Adjust tone for vibrancy. Fix text
and patterns."
}
}

```

Listing 7. Structured Input for Texture Refinement

References

- Yu-Wei Chao, Wei Yang, Yu Xiang, Pavlo Molchanov, Ankur Handa, Jonathan Tremblay, Yashraj S Narang, Karl Van Wyk, Umar Iqbal, Stan Birchfield, et al. 2021. DexYCB: A benchmark for capturing hand grasping of objects. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9044–9053.
- Gheorghe Comanici, Eric Bieber, Mike Schaeckermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261* (2025).
- Shreyas Hampali, Mahdi Rad, Markus Oberweger, and Vincent Lepetit. 2020. Honnotate: A method for 3d annotation of hand and object poses. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 3196–3206.
- Diederik P Kingma. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- Zhengqi Li, Richard Tucker, Forrester Cole, Qianqian Wang, Linyi Jin, Vickie Ye, Angjoo Kanazawa, Aleksander Holynski, and Noah Snaveley. 2025. MegaSAM: Accurate, fast and robust structure and motion from casual dynamic videos. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 10486–10496.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. 2024. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714* (2024).
- Nikhila Ravi, Jeremy Reizenstein, David Novotny, Taylor Gordon, Wan-Yen Lo, Justin Johnson, and Georgia Gkioxari. 2020. Accelerating 3d deep learning with pytorch3d. *arXiv preprint arXiv:2007.08501* (2020).
- SAM 3D Team, Xingyu Chen, Fu-Jen Chu, Pierre Gleize, Kevin J Liang, Alexander Sax, Hao Tang, Weiyao Wang, Michelle Guo, Thibaut Hardin, Xiang Li, Aohan Lin, Jiawei Liu, Ziqi Ma, Anushka Sagar, Bowen Song, Xiaodong Wang, Jianing Yang, Bowen Zhang, Piotr Dollár, Georgia Gkioxari, Matt Feiszli, and Jitendra Malik. 2025. SAM 3D: 3Dfy Anything in Images. (2025). *arXiv:2511.16624 [cs.CV]* <https://arxiv.org/abs/2511.16624>
- Tencent Hunyuan3D Team. 2025a. Hunyuan3D 2.0: Scaling Diffusion Models for High Resolution Textured 3D Assets Generation. *arXiv:2501.12202 [cs.CV]*
- Tencent Hunyuan3D Team. 2025b. Hunyuan3D 2.5: Towards High-Fidelity 3D Assets Generation with Ultimate Details. *arXiv:2506.16504 [cs.CV]* <https://arxiv.org/abs/2506.16504>
- Ruicheng Wang, Sicheng Xu, Cassie Dai, Jianfeng Xiang, Yu Deng, Xin Tong, and Jiaolong Yang. 2025. Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal training supervision. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 5261–5271.
- Bowen Wen, Wei Yang, Jan Kautz, and Stan Birchfield. 2024. Foundationpose: Unified 6d pose estimation and tracking of novel objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 17868–17879.